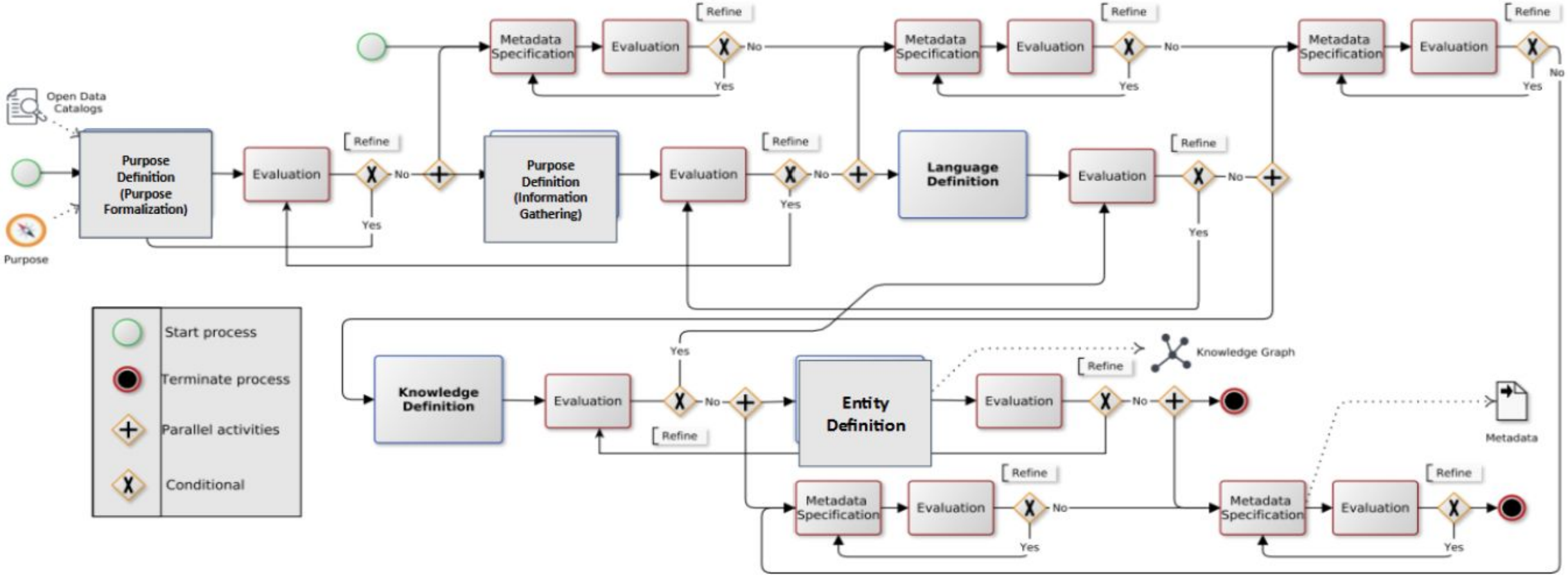
iTelos - Purpose Definition

Fausto Giunchiglia, Mayukh Bagchi

Lecture 15 @ KG 2026

Recap @ *iTelos*

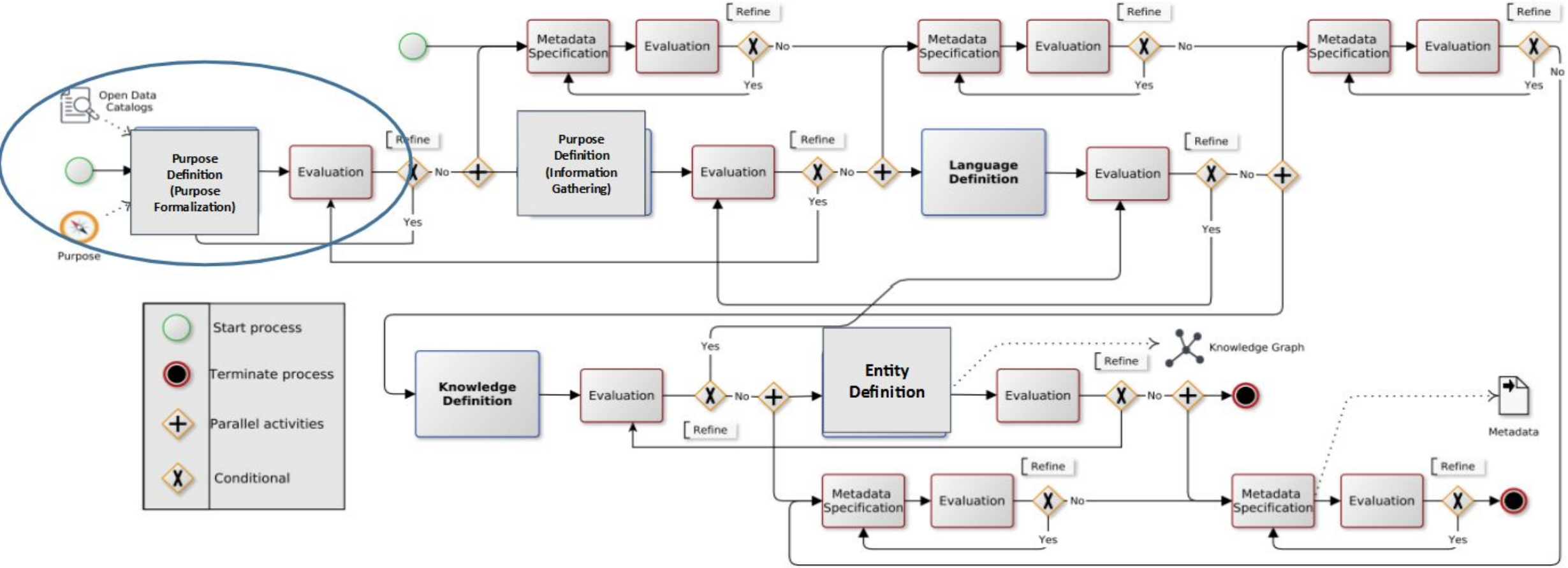


Index of the lecture

- **Objectives of Purpose Definition**

(*) marked slides are for out-of-class consultation.

Purpose formalization @ Purpose Definition



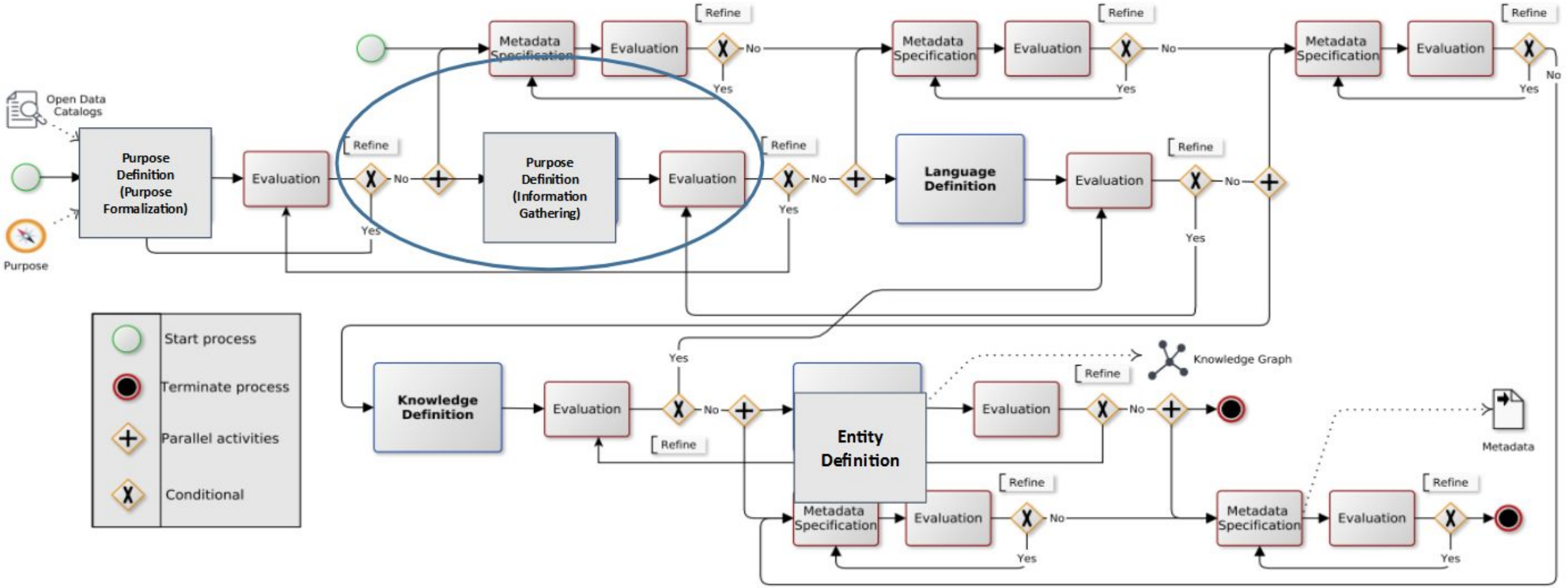
Purpose Definition - *Purpose Formalization* (RECALL)

- The first iTelos phase is focused on the concrete definition of **"The Purpose"**.
- The purpose is **the objective of the user** that is using the methodology.
  - The idea, the why the user is constructing a (or a set of) KG(s).
- It can be seen as **the set of requirements** to be formalized, that will be used to test and validate the iTelos outputs (final and intermediate for each phase).
- However, the purpose implicitly include the "user point of view".
  - **By defining the purpose, the user will also represent her way to represent the real world entities!**

Purpose Definition - *Purpose Formalization (CONTD.)* (RECALL)

- **Input**: a natural language sentence representing the user's Purpose (plus, optionally, a list of already identified data sources to be exploited).
- **Output**: a set of documents and models in which the Purpose's requirements are extracted and formalized.
- **Objective**: to formalize the functional requirements implicitly included in the input user's purpose.

Information Gathering @ Purpose Definition



Purpose Definition - *Information Gathering* (RECALL)

- **Input**: a set of data sources identified previously, plus the formalized user's purpose.
- **Output**: a set of resources collected from the input data sources, suitable to satisfy the purpose.
- **Objective**: this activity of iTelos aims at collecting the resources, to be processed, with the objective to build the final KG(s)

Purpose Definition - *Information Gathering (CONTD.)* (RECALL)

- The gathering of information includes the collection of resources **for all the representation layers**: language, knowledge and data.
- Notice how, depending by the source from which the resources are collected, they have different levels of quality:
  - **Existing sources**: the resources are collected from the existing sources, which not necessarily adopt iTelos, e.g., via open data catalogs, scraping.
  - **KG sources**: the resources collected have been produced by iTelos executions, e.g., via past KG projects, KnowDive sources.

Index of the lecture

- Objectives of Purpose Definition
- **Purpose Formalization**

What is The Purpose ?

- We always act with the constant of our own point of view over ourselves and the world, **"The Purpose"**.
- Our daily activities, at work and in our free time are led by a purpose that somehow **encodes our way to perceive ourselves, the world** in which we live, **and how we interact with it**.
- We use the purpose to **transform our perception into our knowledge**.
- It is important to notice how the purpose **is different for everyone**, not only regarding the ideas/objectives we could have but also considering the way we perceive ourselves and the world around us.
    - **Example**: a PhD student can see the university as my workplace, while a master's student who is also working in a company can see the university as a place to improve the skills required for her job in the company. We are both students, but we perceive ourselves differently as well as different is our perception of the same real world entity (the university, in this case).

Purpose definition (1)

- The purpose is our own way to interpret the external knowledge and transform it into our own.
- The purpose is the primary input of the union process which takes as secondary input what we perceive and produces in output the knowledge defining us.
- From the process (iTelos) point of view, the Purpose is the container of the information requirements to be satisfied by the iTelos final output (the Entity Graph).

Purpose definition (2)

- iTelos defines the Purpose with two versions:
  - **The informal Purpose**: a natural language sentence expressed by the iTelos user (usually the Domain Expert), which state the objective for which the iTelos methodology needs to be applied.
    - **Example**: "I want to build a KG that represents how the environmental impact in Italy has changed over the last five years and what actions in my life have an impact on this."
  - **The formal Purpose**: the first iTelos phase aims a t formalizing the informal purpose, to extract and clearly define the requirements which will be used to lead the entire KGC process in the next phases. The formal Purpose is composed by:
    - a **set of concepts and terms** used to define
    - an **ER model**, representing the first purpose-specific version of the KG's knowledge layer.
    - A set of **Competency Questions (CQs)** which will be turned into the query used to test and evaluate the final Entity Graph.

Purpose Definition - *Objectives*

- It clearly appears that the definition and the understanding of a Purpose is crucial to produce KGs capable of fulfilling it.
- Different KGs can be created, supporting different applications and services, depending on the context expressed by the Purpose.
- The objective of the first iTelos phase aims at extract and formalize the requirements expressed by a Purpose, thus transforming the Informal Purpose into a Formal Purpose.
- This phase requires high precision, because the formal Purpose will be used to lead the entire iTelos process, it will be the key for decision making tasks required for ontology modelling and data engineering activity.

Overall KG Purpose - *example*

The initial, and thus informal, purpose which has been later formalized, is shown below:

> *"A service that will facilitate the finding of schools, including details about the school and courses offered, in the region of Trentino based on city, commune, school type, course duration, and teaching activities schedules."*

From here, we can understand that the project's scope (**Domain of Interest**) includes information regarding the school facilities in Trentino as well as the municipalities of the autonomous region of Trento. Moreover, the courses offered by each educational facility with their associated duration should be included.

This research seeks to provide a complete analysis of the educational facilities in Trentino which comprehend geospatial and temporal domains.

Phase 1 - Purpose Definition - Activities

- **Personas & Scenarios definition**: formalize the context and the actors involved in the project, by describing the types of actors involved as well as the spatial-temporal constraints defining the reference context.
- **Competency Question definition**: state the requirements to be satisfied by the final KG, by defining a list of question that the final Entity Graph should be able to answer.
- **Concepts Identification**: identify the terms and concepts which will be used to define the entity types and the relative properties, to be considered in the final KG.
- **ER modeling**: formalize the Purpose into an ER model, by using the concepts previously identified.

Purpose Definition - Personas & Scenarios (or Use cases) definition

- The goal of this activity is to **formalize the initial purpose statement** received as natural language sentence.
- Such a formalization follow an approach that aims at **extracting the specific information** from the initial Purpose. **The below items must be detailed in the project report document**.

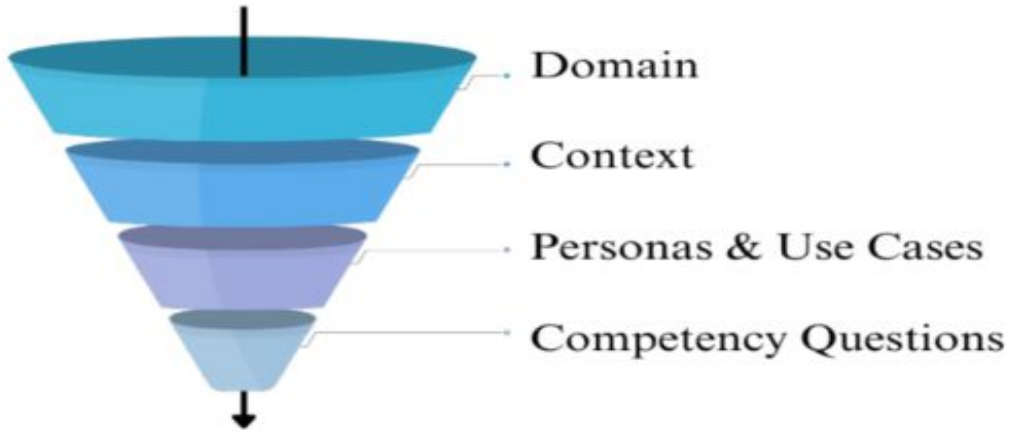


Figure 4: Funnel approach on the Purpose

Purpose Definition - Personas & Scenarios definition

- **Domain of interest**: It refers to the area of knowledge or field of study of interest [36]. Examples are the domains capturing knowledge about daily lives, such as music, tourism, and health, or geographical domains, like Trentino autonomous province.

- **Context**: The second level refers to the context description. The first prescriptive definition of Context referred to it as a location, identities of nearby people and objects, and changes to those objects [37]. More in details the context is defined over three main dimensions:

  - **Geographical boundaries**: Aspects that geographically constrain the problem.
  - **Temporal boundaries**: Aspects that constrain the problem in time.
  - **Domain boundaries**: Domain specific aspect constraining the problem.

---

[36] Fausto Giunchiglia and Biswanath Dutta, Dera: A faceted knowledge organization framework, 2011.

[37] Bill N Schilit and Marvin M Theimer, Disseminating active map information to mobile hosts, IEEE Network 8 (1994), no. 5, 22–32.

Purpose Definition - Personas & Scenarios definition

- **Personas & Scenarios**: user-centered subsets triggered by various subjects, *Personas*, and their real-world perceptions, called Use Cases or *Scenarios*.
  - **Personas** generation is a widely heralded technique that provides semi-fictional subjects characterising the perception and needs of larger groups of end-users.
  - Moreover, **Use Cases** are an essential complement to personas, ensuring a complete and good representation of end-users, by describing the environment in which the personas act.

Purpose Definition - *Persona - Example*

2.2.1 Vincente Trevisan

2.2.1.1 Differentiator
Vincente is a 14 years old boy who lives in Margone, a small village in Trentino. He has just finished middle school and is interested in attending a high school that allows him to become a software engineer. Despite being a native of Trentino, he does not know which nearby high schools provide such a kind of education.

2.2.1.2 Technological Resources
Internet, Smartphone.

2.2.1.3 Needs
Autonomy, Curiosity.

2.2.1.4 Purposes
Vincente wants to find a school where he can learn to be a software developer.



Vincente Trevisan

Personal Information

ID	1
Age	14
Sex	♂
Occupation	Student
Driver	✗
Education	Middle School

2.2.2 Azzurra Gentile

2.2.2.1 Differentiator
Azzurra is a student who lives in Trento. She is completing middle school, where she has done exceptionally well in every subject throughout the year. She truly enjoyed all of the topics in middle school, but she does not have a clue when it comes time to choose which high school to attend. Therefore, she would like to consult a service that gives an overview of all the possible schools available in Trentino to find the right school for her.

2.2.2.2 Technological Resources
Internet, Smartphone.

2.2.2.3 Needs
Competence.

2.2.2.4 Purposes
Azzurra wants to examine different schools' courses and training programs in order to better understand the alternatives for which school to enroll in.



Azzurra Gentile

Personal Information

ID	2
Age	14
Sex	♀
Occupation	Student
Driver	✗
Education	Middle School

2.2.3 Michele Cafaro

2.2.3.1 Differentiator
Michele is a university student pursuing a bachelor's degree in biomedical engineering in Padova. He has started looking for alternative universities to transfer to since he is dissatisfied with the degree he is currently enrolled in. Michele is investigating the option of relocating to the University of Trento for a number of reasons. Therefore, he wonders if the University of Trento would accept the exams he had passed at his own university.

2.2.3.2 Technological Resources
Internet, PC, Smartphone.

2.2.3.3 Needs
Bridging, Autonomy, Curiosity, Competence.

2.2.3.4 Purposes
Michele wants to browse the courses offered by the University of Trento to see if the classes he has previously taken at the University of Padova are offered at the University of Trento, or are at least comparable.



Michele Cafaro

Personal Information

ID	3
Age	20
Sex	♂
Occupation	Student
Driver	✓
Education	High School

2.2.4 Simona Tocci

2.2.4.1 Differentiator
Simona lives in Pergine Valsugana. She is the mother of a child, and now that her child is old enough, she wants to take the child to a kindergarten because she has to work and therefore cannot leave the child at home alone. Since she has a car, she does not need the kindergarten to be strictly close to Pergine Valsugana.

2.2.4.2 Technological Resources
Internet, PC, Smartphone.

2.2.4.3 Needs
Security.

2.2.4.4 Purposes
Simona wants to find a kindergarten for her son.
She also wants to receive feedback from any previous customers, ensuring other parents that a potential kindergarten is a safe and healthy environment for kids.



Simona Tocci

Personal Information

ID	4
Age	34
Sex	♀
Occupation	Worker
Driver	✓
Education	University

Purpose Definition - *Scenario - Example*

2.3.3 Dissatisfaction

2.3.3.1 Description

I want to change schools because I am not happy with my current school. My grades haven't been as good as I want them to be and I feel like I'm not receiving the best education I could get. I'm not sure if it's the school itself or if it's me, but I'm not feeling it. I think it would be a better fit for me to transfer to a different school and see if I can get better grades and have a better education.



Dissatisfaction

Information

Scenario ID 3

2.3.4 Kindergarten for my child

2.3.4.1 Description

My child is 3 years old and he is starting kindergarten this fall. I work hard and I don't always have time to take care of him, so I want him to be safe and happy. I think it would be best for him if he attends a kindergarten where the kids are all from similar backgrounds and where the teachers are experienced and know how to handle kids. I want him to socialize with other kids his age.



Kindergarten

Information

Scenario ID 4

2.3.5 Erasmus student

2.3.5.1 Description

I am a student at a foreign University. I have been interested in studying abroad for a while, and I have decided that I would like to take part in the Erasmus+ program to complete an academic internship at a foreign University in Europe. I found the University of Trento on the Erasmus+ website, and thus I would like to apply to the program.

However, I am not sure if the University of Trento is the right place for me to do my research internship. I am not sure if the research opportunities are good, so I ask one of the professors at the University of Trento if he knows of any opportunities that I could take part in while I am there. The professor tells me that the University of Trento offers a variety of research internships, and I can look into those opportunities if I am interested.



Erasmus student

Information

Scenario ID 5

2.3.6 Best of education

2.3.6.1 Description

I am a parent of a young child and I want my child to attend the best schools possible. I understand the importance of good education and I want to ensure that my child receives the best opportunities possible. I am able to afford to send my child to the best private schools in the area, but I am also open to the idea of sending my child to a public school. I want to make sure that my child has the best chance of achieving their goals and excelling in life.



Best of education

Information

Scenario ID 6

Purpose Definition - Competency Question definition

- The second activity add one more step in the formalization of the initial purpose, by **extracting the KG functional requirements**, from the output of the previous activity, shaping them as **Competency Questions (CQs)**.
- **Competency Questions**: a list of natural language questions. Each question defines a need (a future query to the KG) that should be satisfied by the final KG. Each query refers to a Persona into a specific Scenario.
- **NOTE**: it is important to notice how the definition of CQs is crucial for the design of the final KG.
  - A **poor, set of CQs doesn't provide enough information** regarding which (information) entities needs to be modeled in the KG.
  - A set of CQs with **low diversity, does not represent precisely all the possible (information) details/cases** that the KG should be able to support.
  - **Project Hint**: **Be sure your CQ list is rich of heterogeneous CQs!**

Purpose Definition - *CQ - Example*

Scenarios	Personas	Competency Questions	Common Entities	Core Entities	Contextual Entities
1	1	Give me how many schools have courses on Software Engineering	Institute, School		Study Course
1	1	Give me schools with courses on Software Engineering	Educational Institute, School	Subregional Academic Division	Study Course
1	1	Give me a list of courses at the University of Trento I can consult when I finish High School	Educational Institute, School		Study Course
4	4	Give me Kindergartens near Pergine Valsugana	Educational Institute, School	Subregional Academic Division	
4	4	Give me Kindergartens in Trentino	Educational Institute, School	Subregional Academic Division	
4	4	Give me contact informations about this particular Kindergarten	Educational Institute, School	Institute Contact Information, Subregional Academic Division	
5	5	Give me the list of professors of University of Trento	Educational Institute, School	Professor	
5	5	Give me the courses a professor teaches	Educational Institute, School	Professor	Study Course

Purpose Definition - Concepts Identification (1)

- The third activity is **Concept Identification**.
- Having the CQs listed down, the next activity aims at **extracting the concepts identifying the entity types** (and their properties) to be modeled in the KG.
- To this end a dedicated spreadsheet is adopted, called Purpose Formalization sheet (PFsheet).
- By following the **Middle-out approach**, this activity has to be done considering both the purpose (knowledge layer) and the data sources available (Data layer)

Scenarios	Personas	Competency Questions	Entities	Properties	Focus
1,2,4	2,3,4	1,2,5,6	Bus	color, route_number, capacity,	Contextual
3	1	-	Airplane	type, ownership, registration_number, model	Common
-	1, 2, 4	1,3	Train	type, ownership, capacity, route	Core
1,2,4,5	2,3,4	1,4,5,6,7	Route	path, distance, start_point, end_point, speed_limit, intersections	Contextual
1	2, 4	1,2	Bus stop	location, served_routes	Core
-	1,4	3	Train station	location, served_routes,	Core
3	1	-	Airport	name, location, transporations	Common
5	-	7	Hospital	name, address	Common
5	-	7	Police station	name, address	Common

Purpose Definition - Concepts Identification (2)

- The Concept Identification activity starts to put the focus over the types of entity (ETypes) and properties which will constitute the Entity Graph knowledge layer.
- Each concepts, in this activity is represented by a terms selected from the CQs, or from the input available sources of data.
- Being focused on the CQs and the input data sources, the result of this activity is a list of **purpose-specific** concepts.
- However, one of the iTelos goals is to produce KG not only purpose-specific, but also **reusable and interoperable**.
- To achieve such goal, since the first phase, the concepts are categorized using the **Focus** parameter.

Purpose Definition - Concepts Identification - Focus

- The Focus is a parameter used to indicate how much an EType or a property is "crucial" for a specific purpose.
  - In other words, how much an EType or a property represented by a concept, is required to satisfy the current purpose.
- The Focus and for each EType and property, can have three value:
  - **Common**: general ETypes or properties for the purpose considered.
  - **Core**: specific ETypes or properties for the purpose considered.
  - **Contextual**: very specific ETypes or properties for the purpose considered.
- Clearly, such a categorization has to be done by the Domain Expert, having a full understanding over the Purpose.

Purpose Definition - Concepts Identification - Focus

- The Focus allow **external, future, users to better understand the current project user's Purpose**. In other words, it is a detailed information describing the user point of view.
- Such information allows external users (considering the ongoing project) **to understand how to reuse the resources** produced by iTelos. Which ETypes and properties better characterize the current Purpose.
- Notice how the **Focus express at different levels** (Common, Core and Contextual) **the purpose diversity**.
- Such a **diversity impacts over the reusability and interoperability** of the final Entity Graph.
  - **Example**: a contextual ETypes, being more purpose-specific is probably less interoperable (already existing standard ETypes are less applicable to represent it) respect to a common ETypes for which several representations from existing reference standard ontologies, can be adopted.

Purpose Definition - *Concept Identification Focus - Example*

Scenarios	Personas	Competency Questions	Common Entities	Core Entities	Contextual Entities
1	1	Give me how many schools have courses on Software Engineering	Institute, School		Study Course
1	1	Give me schools with courses on Software Engineering	Educational Institute, School	Subregional Academic Division	Study Course
1	1	Give me a list of courses at the University of Trento I can consult when I finish High School	Educational Institute, School		Study Course
4	4	Give me Kindergartens near Pergine Valsugana	Educational Institute, School	Subregional Academic Division	
4	4	Give me Kindergartens in Trentino	Educational Institute, School	Subregional Academic Division	
4	4	Give me contact informations about this particular Kindergarten	Educational Institute, School	Institute Contact Information, Subregional Academic Division	
5	5	Give me the list of professors of University of Trento	Educational Institute, School	Professor	
5	5	Give me the courses a professor teaches	Educational Institute, School	Professor	Study Course

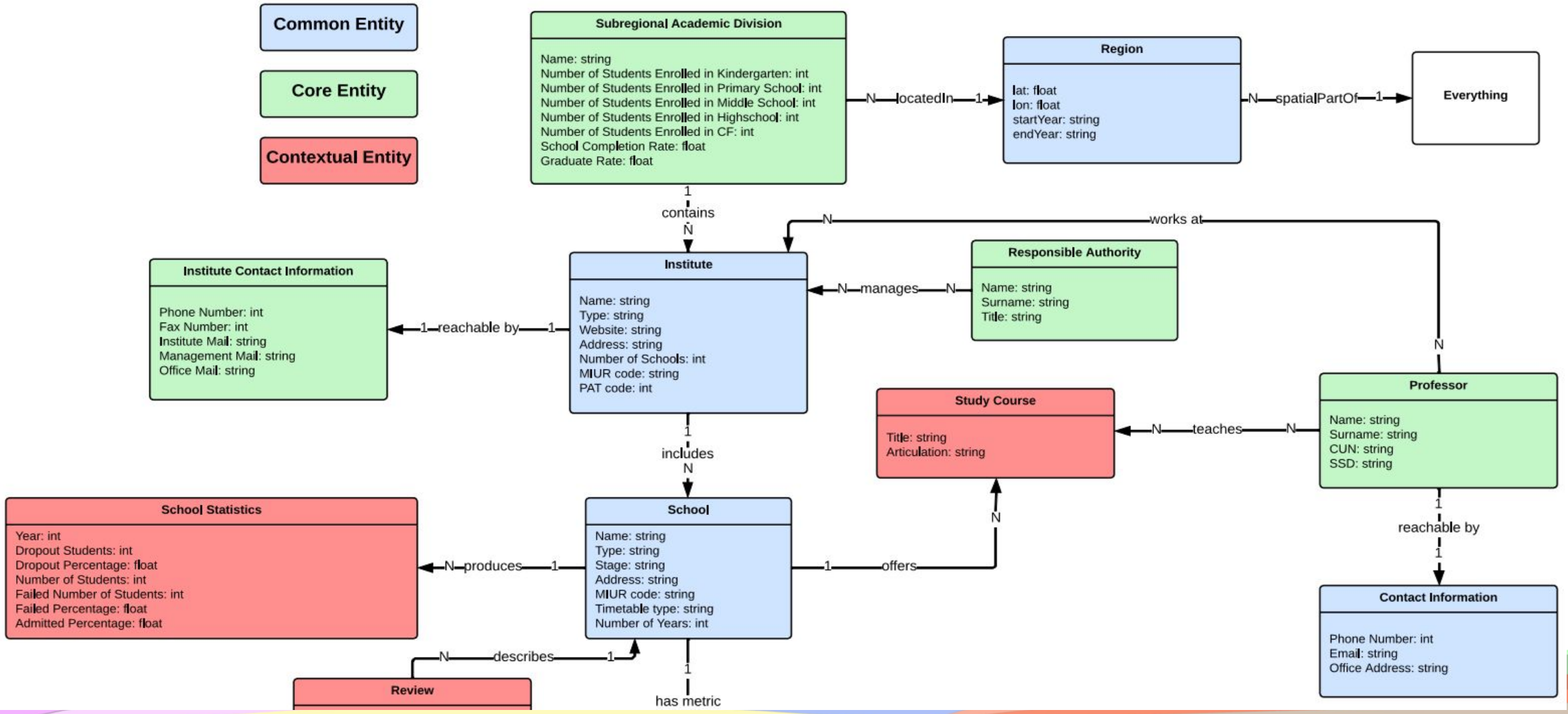
Purpose Definition - ER Modelling

- The fourth activity is **ER modelling**.
- The goal is to **finalize the definition of the formal Purpose**, by modelling an ER model using the concepts, previously selected, representing ETypes and properties.
  - The **ER model** is the first (graphical) version of the final KG structure (or knowledge layer).

ER Model - Definition

- An **Entity–Relationship (ER) Model** describes interrelated things of interest in a specific domain of knowledge.
- It is composed of **classes** / **entity types** (etypes) (which classify the things of interest, i.e. **entities**) and specifies **relationships** that can exist between entities (instances of those entity types).
- The ER model is, thus, an **abstract data model** that defines a data or information structure which can be implemented in a data/knowledge base.
- It is usually drawn in a graphical form as **boxes (classes)** that are connected by **lines (relationships)** which express the associations and dependencies between entities.
- An ER model is the **informal foundation** for the specification of domain-specific teleologies.

Purpose Definition - *ER Model - example*



Index of the lecture

- Objectives of Purpose Definition
- Purpose Formalization
- **Information Gathering**

Purpose Definition - *Information Gathering* [contd.]

The first activity of the current phase, aims at identifying and accessing the sources of information provided as input, and, eventually discovering other sources (if those in input are not sufficient).

Clearly this activity is where the data reuse appears more evident. The user is literally looking for existing data sources providing the information she needs.

As already discussed in the previous lectures, due to the data heterogeneity at information (data source) level, there are multiple types of data sources to be considered.

Purpose Definition - *Information Gathering*

This step is focused on the collection of those resources required for building the final EG, accordingly with the Purpose detailed in the previous phase.

Note - The "resources" mentioned above include:

- Language-level datasets
- Knowledge-level datasets
- Data-level datasets

The Information Gathering phase is not only responsible for the collection of such datasets, it aims also to improve their quality and reusability by cleaning datasets noise adopting well-known standards

Purpose Definition - *Information Gathering* [contd.]

Depending by which source is considered, the collection of datasets requires less or more effort. The data sources provided to you in the KG project proposals have:

- clear policies of data distribution;
- direct distribution/download;
- distribution on demand through request to the data owner (easy communication with data owners).
- High quality metadata, describing the available resources. This increase the findability ad accessibility of the resources distributed.

Purpose Definition - *Information Gathering* [contd.]

You may choose to gather external datasets from external data sources, wherein:

- the data is not always distributed, sometimes it is only published or visualized online.

It is not always clear how to concrete access the datasets. Sometimes the source has to be processed to get the data via:

- asking for datasets directly to owners (if the owner contacts are indicated)
- accessing data through automatic or semi automatic portals/API;
- scraping data from sources (this usually requires scraping libraries customization);
- **via LLMs**

Purpose Definition - *Information Gathering* [contd.]

- Collecting data, in general, aims to achieve the following two results:
  - Increase the number of **entities** and/or entity types.
  - Increase the number of **entity attributes** and/or entity type properties.

- Are the resources collected covering your list of CQs ?
  - **yes** - let's proceed on.
  - **no** - go back to source identification.

Information Gathering - Dataset Cleaning

- The cleaning activity aims to **remove "noise"** from the set of resources collected.
- "Noise" is intended to be:
  - entire **datasets** (could be downloaded as a bunch from a source) without any information required to satisfy the Purpose.
    - It happens often collecting data automatically or receiving huge amount of data.
  - **Etypes and/or entities**, within a single dataset, with no relevance for the Purpose.
  - **Properties and/or attribute**, within single datasets, with no relevance for the Purpose.
  - Any of the above, having **NULL** value.

Information Gathering - Dataset Standardisation (1)

- Now the set of datasets has been finalized.
- The Standardisation activity aims to:
  - **Align the differ formats** present in the heterogeneous dataset (datasets formats and data types), by adopting reusable and interoperable standards.
    - **Example**: For a tabular datasets, the Microsoft Excel format could be ok, but it is a proprietary formats. CSV, being an open format, is more reusable (i.e., for the user that do not use Microsoft apps).
  - **Anonymise the data** collected; required only if sensible information (like personal data) are included in the datasets collected.

Note: the format alignment over common standards (CSV, XML, TSV, JSON, RDF and OWL) is strongly required, mainly for two reasons:

- Reusability, Interoperability.
- Compliance with iTelos process.

UNIVERSITY OF TRENTO - Italy

Dipartimento

Data Cleaning via LLMs

[LINK TO THE THESIS]

POLITECNICO MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE E DELL'INFORMAZIONE

Designing and Evaluating Prompting Strategies for LLM-based Data Preparation

Tesi di Laurea Magistrale in Computer Science and Engineering - Ingegneria Informatica

Thomas Puppinato, 225489

Abstract: The rapid evolution of Large Language Models (LLMs) has reshaped the field of Artificial Intelligence (AI), demonstrating exceptional capabilities in Natural Language Processing (NLP) tasks such as text generation, text classification, machine translation, question answering, and summarization. Moreover, they have also proven capable of performing operations in new domains such as data analysis, problem-solving, and code generation. LLMs, thanks to their ability to interpret instructions, infer context, and generate coherent outputs, represent promising tools for data-driven workflows. Among these workflows, data preparation is one of the most critical and complex phases, which encompasses a collection of operations aimed at improving data quality. Given the expanding capabilities of LLMs beyond traditional NLP tasks, their potential to detect errors, transform data, and identify dependencies has made the question of whether they can support data preparation an important research direction. However, their actual effectiveness in these tasks remains uncertain. This thesis therefore investigates whether LLMs can be effectively used to support data-preparation pipelines. Moreover, since LLMs often struggle with complex tasks and can benefit from additional guidance to improve accuracy, various prompting techniques have been developed to guide them toward more relevant responses. To this aim, we design and evaluate different prompting strategies to enhance LLM performance in executing data preparation tasks.
To achieve this goal, an experimental framework was designed to evaluate the capabilities of three state-of-the-art LLMs in performing data preparation on structured datasets characterized by different degrees of complexity and noise. We focused on data exploration tasks, such as data profiling and dependency discovery, and cleaning tasks, such as data wrangling, data standardization, outlier detection, data imputation, and data de-duplication. Different prompting strategies were designed to enhance models' ability to perform such demanding tasks. Model outputs were evaluated through a set of customized metrics tailored to users' needs in being supported while performing data preparation.
Results demonstrate that effective prompt design can significantly enhance LLM capabilities in supporting data-preparation tasks, although such improvements are not observed across all operations.

Advisor:
Prof. Cinzia Cappiello

Co-advisors:
Camilla Sancricca

Academic year:
2024-2025

Key-words: Large Language Model; Data preparation; Data quality

Data Cleaning via LLMs

LINK TO THE THESIS

5.4.1 Zero-shot prompt

A zero-shot prompt was used to extract a baseline performance of the LLMs, that is, the level of results achievable without applying any advanced prompting techniques or additional guidance.
Its structure is the following:

```
Consider this dataset:
{{csv_text}}
Can you do {{task}} on it?
```

where:

- `{{csv_text}}` is the textual representation of the .csv file containing the dataset.
- `{{task}}` is the name of the analyzed data preparation task.

5.4.2 Detailed zero-shot prompt

The detailed zero-shot prompt shares the same content as the zero-shot prompt; however, it subsequently includes a series of specific operations designed to guide the model more explicitly through the execution of the task.
The prompt is:

```
Consider this dataset:
{{csv_text}}
Can you do {{task}} on it?
To complete this task, you need to follow
these steps:
{{list_of_steps}}
```

where `{{list_of_steps}}` is the list of instructions to be performed, which varies depending on the specific task.

```
Consider this dataset:
{{example_dataset_1}}
{{example_output_1}}
Consider this dataset:
{{example_dataset_2}}
{{example_output_2}}
Consider this dataset:
{{csv_text}}
Can you perform {{sub-task}} on it?
```

where:

- `{{example_dataset_1}}` and `{{example_dataset_2}}` are small datasets containing a limited number of rows (ranging from five to ten) which are characterized by the presence of errors. The type of pollution depends on the nature of the task to be performed.
- `{{example_output_1}}` and `{{example_output_2}}` are the corresponding outputs obtained after executing a specified sub-task on `{{example_dataset_1}}` and `{{example_dataset_2}}`.
- `{{sub-task}}` is the sub-task to be performed on the dataset.

Task
Data Profiling
Data Cleaning
Outlier Detection
Data Standardization
Data Imputation
Data Deduplication
Data Wrangling
Dependency Discovery

Three of the most commonly used LLMs in the literature were selected, all of which demonstrate high performance on tasks typically performed in widely used benchmarks[1]: DeepSeek from DeepSeek, Gemini from Google, and GPT from OpenAI. In a preliminary phase of the experimentation, the prompts were tested against the following models:

- DeepSeek: `DeepSeek-V3-0324`
- Gemini: `gemini-1.5-pro`
- GPT: `gpt-4o-2024-08-06`

Identifying Attributes Definition (Informal) - *Example*

Identifying Attributes are informally derived from Data Sets.

Definition: Identifying attributes are the unique data elements that distinguish and disambiguate one instance of an entity type from another, observed directly from how data records are structured in datasets.

Example (EType: School):
Dataset → Columns like School_ID, School_Name, Location, Type, Principal_Name
Identifying Attribute → School_ID (uniquely identifies each school)

In short: Identifying attributes = data fields that provide unique identity to each entity instance.

Descriptive Attributes Definition (Informal) - *Example*

Descriptive Attributes are informally derived from Competency Questions (CQs).

Definition: Descriptive attributes are the key facts or characteristics that describe an entity type, identified by what the CQs ask about it.

Example (EType: School):
CQs → What is the name of the school? Where is it located? When was it established?
Descriptive Attributes → Name, Location, Establishment Year, Type, Principal

In short: Attributes = direct answers sought about the entity itself, not its relationships.

Purpose of Datasets - *Example*

The purpose of data sets gathered should be in relation to an Entity Type and its Properties from CQs

Data sets provide the actual recorded instances of an entity type and its properties, serving as the empirical source that validates or enriches what is defined through Competency Questions (CQs).

Example (EType: School):
CQs → define what information is needed (Name, Location, Establishment Year, Type, Principal)
Data Set → provides real data values for these properties across multiple schools.

In short: CQs define what to capture; Data Sets show how it exists in reality.

Identifying Attributes Definition (Informal) - *Example*

Etype	Primary	Secondary
Geographic Region	name of the region	
Subregional Academic Division	name of the municipality	
Institute	MIUR code	integer id
School	an integer id	
Institute Contact Information	integer id	
Invalsi	municipality code, school stage, school year and subject	
Professor	UID assigned by Università Digitale	
Study course	UID assigned by Università Digitale	integer identifier
Review	integer id	
School Statistics	integer id	
Education Quality	integer id	

Information Gathering - Dataset Standardisation (2)

The best practices for the generation of highly reusable and interoperable data are provided by:

- FAIR Principles
- 5-Stars Open Data

Purpose Definition - *Information Gathering - example*

3.3.1.3 educational_institutions.csv

educational_institutions.csv

Attributes	Description
institute_name	name of the institute
institute_phone	mechanical code
institute_fax	description of the course
institute_website	website of the course

3.3.1.4 responsible_person_educational_institutions.csv

responsible_person_educational_institutions

Attributes	Description
institute_name	name of the institute
responsible_person	name of the responsible person
responsible_person_role	role of the responsible person

3.3.1.5 schools_in_trentino.csv

schools_in_trentino

Attributes	Description
institute_name	name of the institute
school_email	email of the school
school_type	whether the school is public, private, religious or secular
school_responsible_person_name	name of the school responsible person's role
school_name	name of the school

3.3.11 unitn_courses.csv

unitn_courses

Attributes	Description
id_course	course's id
course_name	name of the course
degree_program	degree in which the course is registered to

3.3.12 unitn_staff.csv

unitn_staff

Attributes	Description
id_employee	employee's id
name	name of the employee
surname	surname of the employee
CUN area	CUN area of the employee (could be missing if the person is not a professor)
SSD	disciplinary scientific sector of the employee (could be missing if the person is not a professor)

3.3.13 departments.csv

departments

Attributes	Description
id_department	id of the department
name	name of the department

Index of the lecture

- Objectives of Purpose Definition
- Purpose Formalization
- Information Gathering
- **Purpose Definition @ Oncovirus KG'25 Project**

UNIVERSITY
OF TRENTO - Italy

Purpose @ Oncovirus KG

2 Purpose Definition

2.1 Informal Purpose

The purpose of this KG is to establish a unified and semantically rich representation of the biological mechanisms through which viruses contribute to cancer development. It seeks to integrate dispersed information about oncoviruses, their encoded proteins, and the host cellular components they interact with, in order to elucidate the molecular and systemic connections that underlie viral oncogenesis.

By connecting entities such as viruses, host and viral proteins, cellular pathways, phenotypes, cancers, and vaccines, the graph will enable researchers to navigate complex relationships that are otherwise scattered across literature and databases. The goal is to create a framework that not only supports data integration but also facilitates reasoning, discovery, and hypothesis generation.

This KG will serve as a bridge between molecular biology, computational modeling, and clinical understanding, allowing scientists to trace how a virus infects a host, manipulates its molecular machinery, disrupts cellular pathways, and ultimately drives tumorigenesis. Furthermore, it will provide a foundation for vaccine and therapeutic research by highlighting targets and pathways implicated in virus-induced cancers.

2.2 Domain of Interest

The domain of interest encompasses the molecular, cellular, and clinical dimensions of virus-induced cancers, focusing on the interactions between oncogenic viruses, host cellular machinery, cancer-related pathways, and therapeutics. The KG integrates biological, molecular, and clinical information to model how viruses contribute to tumorigenesis across different organisms and tissue types.

UNIVERSITY OF TRENTO - Italy

Purpose @ Oncovirus KG (contd.)

2.2.1 Space

The spatial scope of this study primarily involves human biological systems, where the focus lies on host–virus interactions at the molecular and cellular levels. Specifically, the graph captures protein interactions within cellular compartments and within tissues or organs that serve as the primary sites of infection and cancer development (e.g., liver for HBV/HCV, cervix for HPV, lymphatic tissue for EBV).

The viruses represented include major oncoviruses of clinical relevance, such as Human Papillomavirus (HPV), Hepatitis B Virus (HBV), Hepatitis C Virus (HCV), Epstein–Barr Virus (EBV), and Human T-lymphotropic Virus (HTLV-1), reflecting a global distribution across diverse populations and cancer types.

2.2.2 Time

The temporal scope extends from the initial viral infection to the development and progression of virus-induced cancers. This includes stages such as viral entry, genomic integration, latency, immune evasion, cellular transformation, and tumor formation.

While the KG is not bound to specific chronological datasets, it integrates findings from studies conducted between the years 2000 and 2025, encompassing both historical discoveries and contemporary research in virology, oncology, and bioinformatics.

UNIVERSITY OF TRENTO - Italy

Purpose @ Oncovirus KG (contd.)

2.3 Scenarios

These represent practical use cases demonstrating how the KG can be used.

1. **Identify Virus–Cancer Links:**
   A researcher queries which viruses are causally associated with specific cancer types (e.g., HPV → cervical cancer).

2. **Map Virus–Host Interactions:**
   A bioinformatician explores which human (host) proteins interact with viral proteins to identify potential therapeutic targets.

3. **Analyze Disrupted Pathways:**
   A molecular biologist traces which signaling pathways are affected by viral proteins to understand oncogenic mechanisms.

4. **Explore Vaccine Coverage:**
   A public health analyst checks which viruses have vaccines and which remain without preventive measures.

5. **Phenotype-based Discovery:**
   A virologist studies virus phenotypes and correlates them with cancer incidence or severity.

6. **Functional Context of Proteins:**
   A computational scientist investigates how host and viral protein functions/localizations relate

UNIVERSITY OF TRENTO - Italy

Purpose @ Oncovirus KG (contd.)

2.4 Personas

Personas represent typical users of the KG and their goals.

1. **Dr. Elisa Marino – Virologist**
   Focus: Understanding viral protein functions and phenotypes.
   Uses KG to connect viral mutations with oncogenic potential.

2. **Dr. James Liu – Computational Biologist**
   Focus: Integrating datasets and analyzing protein–protein interaction networks.
   Uses KG to run queries linking virus-host interactions with pathway disruptions.

3. **Dr. Maria Rossi – Oncologist**
   Focus: Translating molecular data into clinical understanding.
   Uses KG to explore which viral infections are associated with specific cancers.

4. **Dr. Daniel Ortega – Vaccine Researcher**
   Focus: Designing or improving vaccines targeting oncoviruses.
   Uses KG to identify viral proteins that are ideal immunogenic candidates.

5. **Dr. Anna Becker – Molecular Pathologist**
   Focus: Examining cellular pathways impacted by viral infections in tumor samples.
   Uses KG to cross-reference altered pathways and cancer data.

6. **Prof. Samuel Green – Bioinformatics Educator**
   Focus: Teaching integrative omics and KG technologies.
   Uses KG as a pedagogical example of biological knowledge modeling.

7. **Dr. Rania Al-Hassan – Data Scientist**
   Focus: Developing machine learning models on biological graphs.
   Uses KG embeddings to predict novel virus–cancer associations.

UNIVERSITY
OF TRENTO - Italy

2.5 Competency Questions (CQs)

These questions define what the KG must be able to answer.

1. **Virus–Cancer Relationships**
   Which viruses are known to cause specific types of cancer?

Purpose @ Oncovirus KG (contd.)

2. **Protein Interactions**
   Which host proteins interact with specific viral proteins, and what is their biological function?

3. **Pathway Involvement**
   Which pathways involve the host proteins that interact with viral proteins?

4. **Vaccine Coverage**
   Are there existing vaccines targeting a given oncovirus?

5. **Viral Phenotypes**
   What phenotypic traits (e.g., virulence, tropism) are associated with a specific virus?

6. **Virus Targeting and Cancer Association**
   What viruses target a specific host protein, and what cancers are they associated with?

7. **Pathway Disruption by Viral Proteins**
   Which viral proteins disrupt a specific host biological pathway?

UNIVERSITY OF TRENTO - Italy

Purpose @ Oncovirus KG (contd.)

2.6 Concepts Identification

2.6.1 Entity Types (ETypes) and Attributes

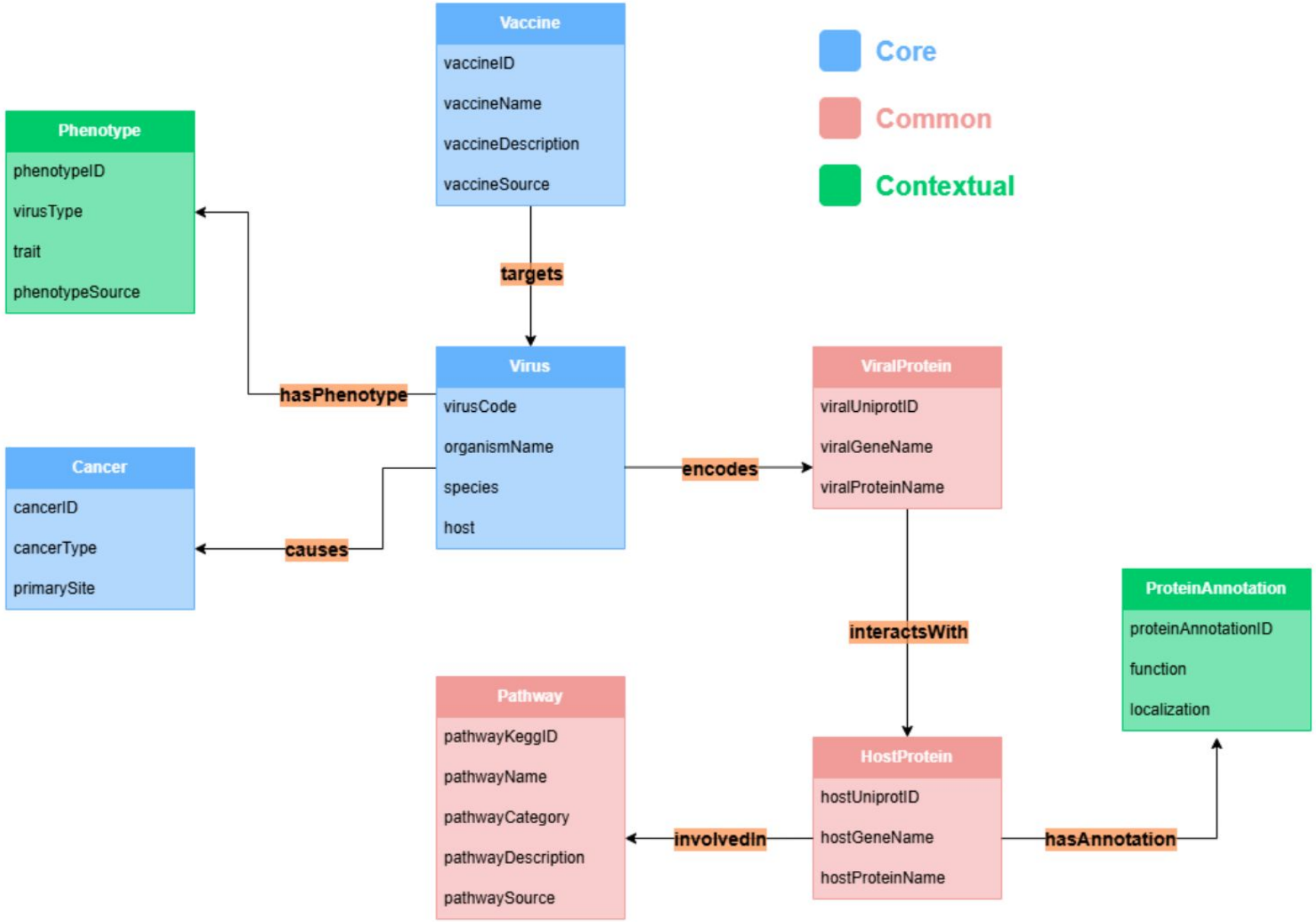
Entity Types	Categories	Description / Role in KG	Attributes
Virus	Core	Represents oncogenic viruses responsible for specific cancers.	virusCode, organismName, species, host
Cancer	Core	Represents cancers associated with viral infections.	cancerID, cancerType, primarySite
Vaccine	Core	Represents preventive vaccines targeting oncoviruses.	vaccineID, vaccineName, vaccineDescription, vaccineSource
ViralProtein	Common	Encoded proteins from oncoviruses that interact with host proteins.	viralUniprotID, viralGeneName, viralProteinName
HostProtein	Common	Human proteins targeted or affected by viral proteins.	hostUniprotID, hostGeneName, hostProteinName
Pathway	Common	Biological pathway or process involving host proteins.	pathwayKeggID, pathwayName, pathwayCategory, pathwayDescription, pathwaySource
ProteinAnnotation	Contextual	Stores metadata describing protein-specific properties (function, localization,	proteinAnnotationID, function, localization

Purpose @ Oncovirus KG (contd.)

2.6.2 Relationships

- Vaccine **targets** → Virus
- Virus **causes** → Cancer
- Virus **hasPhenotype** → Phenotype
- Virus **encodes** → ViralProtein
- ViralProtein **interactsWith** → HostProtein
- HostProtein **hasAnnotation** → ProteinAnnotation
- HostProtein **involvedIn** → Pathway

Purpose @ Oncovirus KG (contd.)



Purpose @ Oncovirus KG (contd.)

2.7.1 EType Categories

We define the following ETypes and categorize them as Common, Core, or Contextual, following the same principles used in our domain analysis:

• **Common ETypes:**

1. **ViralProtein:** represents proteins encoded by oncoviruses that can directly or indirectly interact with host proteins. This EType is fundamental for capturing the molecular mechanisms through which viruses modulate host cellular pathways.
2. **HostProtein:** represents human proteins targeted or affected by viral proteins. This EType provides the necessary foundation to describe virus–host interactions at the molecular level.
3. **Pathway:** represents biological pathways or molecular processes involving host proteins. It enables grouping of host proteins into functional contexts, facilitating higher-level reasoning about disease mechanisms.

Purpose @ Oncovirus KG (contd.)

- **Core ETypes:**

1. **Virus:** represents oncogenic viruses responsible for specific cancers. This EType serves as a primary node in the KG, connecting to multiple other entities (cancers, vaccines, phenotypes and viral proteins).
2. **Cancer:** represents types of cancers associated with viral infections. It allows linking molecular mechanisms to clinical outcomes and supports CQs about oncogenesis and disease prevalence.
3. **Vaccine:** represents preventive therapeutics developed to target or neutralize specific oncoviruses. This EType helps address CQs related to prevention and immunization strategies.

- **Contextual ETypes:**

1. **ProteinAnnotation:** stores metadata describing protein-specific properties such as function, subcellular localization, and evidence type. This EType enriches the molecular host entities with additional biological meaning.
2. **Phenotype:** represents observable virus-related traits such as virulence, cell tropism, or infectivity. This EType is crucial for contextualizing viral behavior and linking it to pathogenic potential.

2.7.2 EType Relationships

We identify a set of biologically meaningful relationships capturing causation, interaction, and participation patterns within the system. Some are straightforward associative links (e.g., "Vaccine targets Virus"), while others represent complex molecular associations (e.g., "ViralProtein interactsWith HostProtein"). These edges are central to addressing the CQs concerning oncogenic mechanisms, molecular targets, and preventive interventions.

Purpose @ Oncovirus KG (contd.)

2.8 Qualitative Evaluation

During this initial phase, the informal purpose definition underwent a minor refinement following the construction of the preliminary ER model. This adjustment ensured a clearer alignment between the conceptual intent of the KG and the structural representation of the domain. The refinement helped articulate the objectives more precisely and confirmed that the emerging model accurately reflected the intended scope and analytical goals of the project.

Purpose @ Oncovirus KG (contd.)

3 Information Gathering

3.1 Data Sources

All the data necessary for this study are derived from multiple established bioinformatics and biomedical repositories [5][6][7][8][9][10]. These sources collectively provide comprehensive information on oncogenic viruses, including Hepatitis B virus (HBV), Human herpesvirus 8 (HHV-8), Epstein–Barr virus, B95 strain (EBV), Human papillomavirus strains 16 and 18 (HPV16, HPV18), and Human T-cell lymphotropic virus strains 1A and 1C (HTLV-1A, HTLV-1C), the viral proteins they produce, and their interactions with human host proteins. The following databases were integrated to construct the final dataset:

3.1.1 NCBI Virus Database

This dataset provides gene annotations, coding sequences (CDS), and protein products, serving as a foundation for linking viral genes to their protein functions and host targets [5].

3.1.2 UniProt Database

Supplies detailed protein-level information, including molecular functions, biological processes, and subcellular localization. Both viral and human host proteins are retrieved from UniProt to ensure consistent identifier mapping and functional annotation [6].

3.1.3 IntAct Molecular Interaction Database

Purpose @ Oncovirus KG (contd.)

3.2 Data Preprocessing

During the data preprocessing phase, interaction information was collected for seven viral entities, derived from five major oncogenic viruses: HBV, HHV-8, EBV (B95 strain), and multiple strains of HPV (HPV16 and HPV18) and HTLV (HTLV-1A and HTLV-1C). Virus–host PPI data were retrieved from the IntAct Molecular Interaction Database. All extraction, merging, and compilation steps were automated using customized Python scripts to ensure efficiency, reproducibility, and comprehensive collection of interaction records. There was no misalignment in the data-gathering process. Following the generation of the fully merged IntAct dataset, an extensive data-cleaning procedure was applied to improve accuracy and ensure consistent annotation. In this step, 2,360 interaction records were removed due to missing required fields, and an additional 148 records were discarded as duplicates, defined as entries sharing the same combination of virus code, viral UniProt ID, and host UniProt ID. Only complete, unique, and biologically relevant interactions were retained for further analysis.

Following this cleanup, all remaining viral and host protein identifiers were cross-referenced with the UniProt database to retrieve detailed protein-level information, including functional annotations, sequence identifiers, and associated biological roles. Entries lacking sufficient UniProt information were excluded, resulting in the removal of 4 viral proteins and 82 host proteins. The final interaction dataset obtained after these filtering steps contained 31 distinct viral proteins and 759 human host proteins, representing a curated and well-validated set of interactions.

Subsequently, each host protein was mapped to its associated biological pathways using the KEGG and UniProt APIs in Python-based annotation workflows. The resulting pathway dataset underwent an additional round of cleaning to remove redundant, ambiguous, or incomplete pathway entries. Before this cleaning step, the pathway dataset consisted of 2,892 records; after filtering, 2,272 high-quality records remained, with 620 entries removed due to missing information or lack of clear annotation.

All final, cleaned, and biologically validated datasets have been saved in the repository as in the following table:

File	Content	Related EType
cancer_data.tsv	cancerID, cancerType, primarySite	Cancer
host_protein_data.tsv	hostUniprotID, hostGeneName, hostProteinName	HostProtein
viral_protein_data.tsv	virusCode, viralUniprotID, hostUniprotID, viralGeneName, viralProteinName	ViralProtein

Purpose @ Oncovirus KG (contd.)

The GitHub repository contains several Python scripts used during the preprocessing and annotation of virus–host protein interactions:

1. **intact_datasets_merger.py**: Merging of virus–host PPI records from the IntAct Molecular Interaction Database; producing a single dataset.
2. **ppi_information_extractor.py**: Automates the extraction of PPI information from the fully merged intact dataset and excluding any duplicated records or ones with missing data.
3. **viral_and_host_protein_extractor.py**: Extracts and compiles the final set of viral and host proteins for downstream analysis, ensuring that only complete, unique, and biologically relevant proteins are included.
4. **protein_annotation_extractor.py**: Retrieves detailed protein-level information for host proteins by cross-referencing UniProt. It collects functional annotations, sequence identifiers, and associated biological roles.
5. **pathway_information_extractor.py**: Maps host proteins to their associated KEGG biological pathways via the KEGG and UniProt APIs.

3.3 Qualitative Evaluation

Throughout the information gathering phase, the ER model was revisited and refined to ensure coherence with the collected datasets. Several attributes that were no longer meaningful or relevant were removed, while others were renamed to improve clarity and consistency with biological terminology and data sources. Although certain relationship labels between ETypes were adjusted for precision, all previously defined relationships were retained, indicating that the original structural assumptions remained valid. Importantly, the informal purpose definition required no modification, demonstrating that the scope and objectives established in the purpose definition phase continued to align well with the data and domain understanding gained during this phase.

4 Schema and Ontologies

The cleaned datasets contain a range of entities and properties essential for analyzing oncoviruses and the interactions between viral and host proteins that disrupt cellular pathways leading to cancer. While most ETypes do not have suitable representations in *Schema.org* or *Bioschemas.org*, we were able to identify corresponding concepts for many of them in *NCBO BioPortal* under specialized biomedical ontologies [11][17][18]. When applicable, these ETypes are aligned with the established ontology terms; otherwise, we rely on schema derived directly from the original datasets.

1. Cancer *(from the Human Disease Ontology, DOID:162 via NCBO BioPortal)*

- **cancerID** (Integer, from the original data): Unique identifier for the cancer entry.
- **cancerType** (Text, from the original data): Category or type of cancer.

Purpose @ Oncovirus KG (contd.)

- **primarySite** (Text, from the original data): Anatomical site or primary location associated with the cancer type.

2. HostProtein *(from the original data)*

- **hostUniprotID** (Text, from the original data): Unique reference to the UniProt Accession Number of the host protein.
- **hostGeneName** (Text, from the original data): Gene name of the host protein.
- **hostProteinName** (Text, from the original data): Protein name of the host protein.

3. ViralProtein *(from the original data)*

- **viralUniprotID** (Text, from the original data): Unique reference to the UniProt Accession Number of the viral protein.

Index of the lecture

- Objectives of Purpose Definition
- Purpose Formalization
- Information Gathering
- Purpose Definition @ Oncovirus KG'25 Project
- **Purpose Definition @ Territory+Health KG'25 Project**

Purpose @ Territory KG

The primary goal of this project is to build a **Knowledge Graph (KG)** that stores, organizes, and provides easy access to information about **essential geographical features and various healthcare facilities** within a specific territory, namely the **Trentino region**. The KG will function as a centralized knowledge hub, connecting citizens, public health planners, and emergency services to critical information regarding the **location, specialization, availability, and accessibility** of healthcare resources across the region.

The **Domain of Interest (DoI)** for the "Territory and Health Facilities" project is the **Trentino region**, focusing specifically on the relationship between its **geographical features** and **healthcare infrastructure**.

The DoI encompasses the following key aspects:

- **Geographical Scope:** The Trentino region (e.g., municipalities, towns, and locations) is the territory under study.
- **Target Entities:** Essential geographical features and different healthcare facilities (hospitals, clinics, specialized centers, emergency medical facilities) within that territory.
- **Project Focus:** Modeling the location, specialization, availability, and, crucially, the accessibility/proximity of these facilities relative to geographical points (towns, given locations).

Purpose @ Territory KG (contd.)

3.1 Purpose Formalization

3.1.1 Scenarios Definition

In this section, a set of usage scenarios, describing the multiple aspects considered by the project purpose (linking geographical features to healthcare infrastructure), are presented.

1. **Emergency Proximity A tourist**, Marco, is hiking in a remote area (he only knows his GPS coordinates). He has a minor emergency and needs to find the absolute nearest hospital and pharmacy, regardless of the municipality.
2. **Local Specialized Care Access** A caregiver, Giulia, lives in the municipality of 'Cles'. She needs to find a semi-residential facility for her elderly father. She wants to find all options within her municipality first, and then expand her search to nearby municipalities if necessary.
3. **Service Gap & Distribution Analysis** A public health analyst, Dr. Rossi, is tasked with identifying potential "service deserts." She needs to map the locations of all psychiatric services to see which municipalities have no local access.
4. **Territorial Capacity Planning** A public health planner, Luca, is reviewing the total capacity for long-term care. He needs to know the total number of beds available in residential facilities for the elderly, aggregated by geographical area (e.g., all facilities in the municipality of 'Trento').

Purpose @ Territory KG (contd.)

3.1.2 Personas

1. **Marco Bianchi** is a 32-year-old tourist visiting Trentino for a hiking trip. His main concern is safety in unfamiliar territory. He needs to be able to find the nearest hospital or pharmacy based on his phone's GPS coordinates in case of an emergency, rather than searching by a town name he doesn't know.
2. **Giulia Verdi** is a 52-year-old resident of Cles. Her main task is finding a suitable care facility for her elderly father. She needs to find all residential or semi-residential facilities that specifically offer Elderly Assistance within her home municipality.
3. **Dr. Elena Rossi** is a 45-year-old public health analyst. Her main task is creating reports on healthcare access and equity. She needs to identify which municipalities do not have a local facility offering specialized services like Psychiatric Assistance to map service deserts.
4. **Luca Verdi** is a 38-year-old health system planner. His main task is managing resource allocation for the upcoming year. He needs to know the total number of beds available in residential facilities aggregated by municipality to assess regional capacity.

Purpose @ Territory KG (contd.)

3.1.3 Competency Questions (CQs)

Initally, we came up with some scratch competency questions that were not based on actual data, but an imaginary idea of what the datasets and data could include. After finalizing the datasets chosen for this project, the questions were either modified or deleted to fit the datasets better, this included an improved set of competency questions that are more precise and helpful. (make more formal, questions include emergency details that got terminated later, usage of ontological aspects)

Table 1: *Competency Questions (CQs)*

Person	No.	Question
Marco Bianchi	1.1	What is the nearest hospital to a current GPS coordinates?
Marco Bianchi	1.2	Find the address and phone number of a specific health structure.
Marco Bianchi	1.3	List all pharmacies within a 5km radius of a specific health facility.
Giulia Verdi	2.1	Which residential facilities for Elderly Assistance are located in a specific municipality?
Giulia Verdi	2.2	Are there any semi-residential structures in a specific municipality? If not, what are the closest ones?
Giulia Verdi	2.3	What specialized services are available at the health structures in a specific municipality?
Dr. Elena Rossi	3.1	Which municipalities do not have any health structure offering Psychiatric Assistance?
Dr. Elena Rossi	3.2	Generate a list of all municipalities that have more than 5 pharmacies.
Dr. Elena Rossi	3.3	What types of hospitals are present in comparison between different municipalities?
Luca Verdi	4.1	What is the total number of beds for elderly care in a municipality?
Luca Verdi	4.2	What is the total capacity of all semi-residential structures in the province?
Luca Verdi	4.3	How many health structures are managed directly by the health service in a specific municipality?

Purpose @ Territory KG (contd.)

Table 2: *Concept Identification: Etypes and Attributes*

Entity Type	Category	Role	Attributes
HealthFacility	Core	Represents the central concept of any physical location providing healthcare services.	StructureCode (ID), Name, Address, CAP, Phone, Email, Website, Lat/Long, ServiceName
Hospital	Core	Represents a specialized facility for comprehensive medical care.	Inherits HealthFacility + ASL_Code, ManagementType
Dispensary	Core	Represents a facility for dispensing medication.	IVA (VAT), DispensaryType (Pharmacy/Parapharmacy)
CareStructure	Core	Represents a facility for residential or semi-residential care.	NumBeds, NumUsers, NumAdmissions, CareType, ManagementType
RehabStructure	Core	Specialized facility for therapeutic and rehabilitation services.	ASL_Code
Municipality	Common	Represents the administrative territory.	MmunicipalityCode (ID), Name, Province

Relationships

The semantic relationships define how the identified entities interact to form the Knowledge Graph structure. These relationships are categorized into inheritance (class hierarchy) and semantic links (object properties).

Inheritance

These relationships define the specialization hierarchy, where specific facility types inherit attributes from the central `HealthFacility` class.

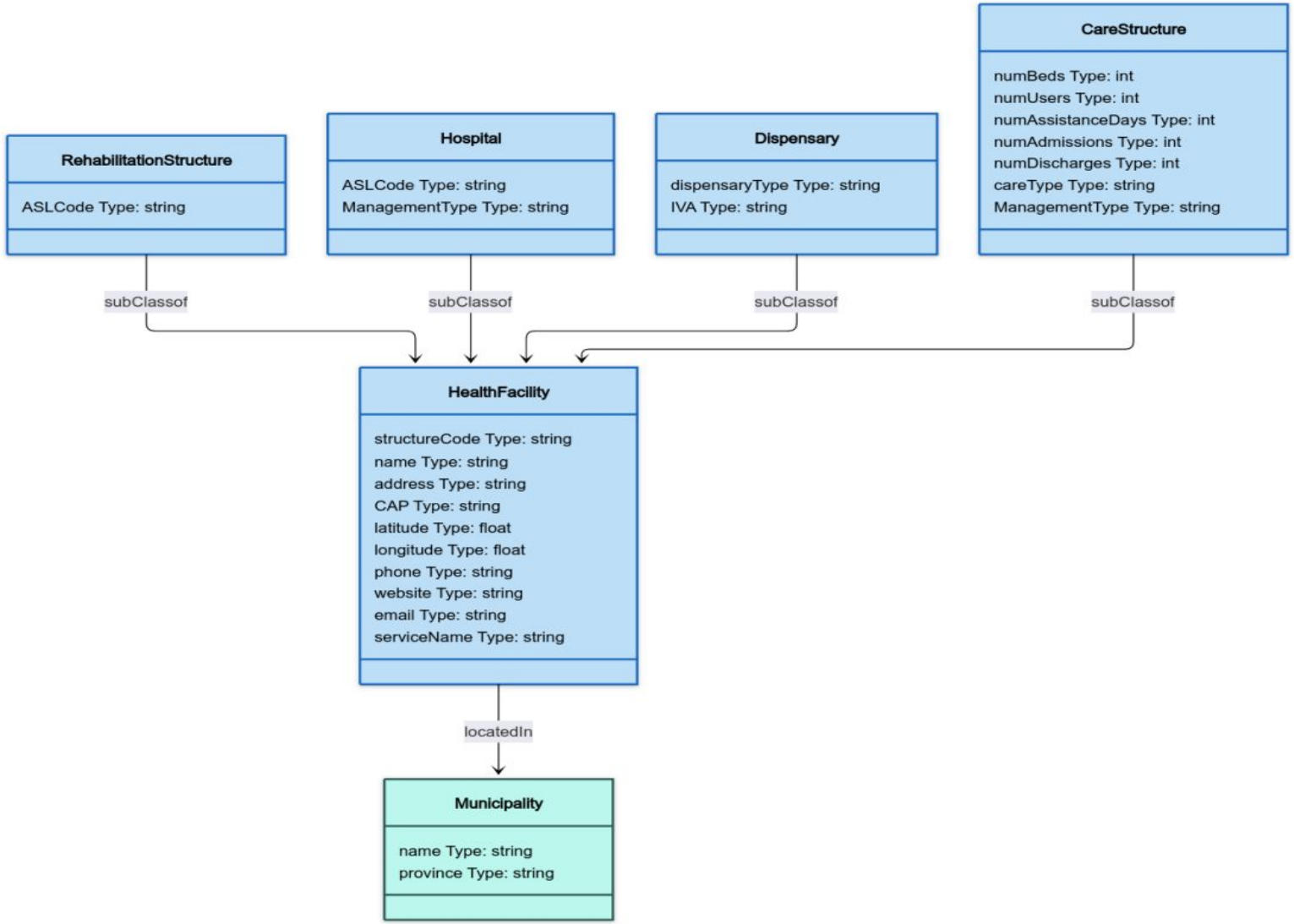
- **Hospital** → `subClassOf` → **HealthFacility**
- **Dispensary** → `subClassOf` → **HealthFacility**
- **CareStructure** → `subClassOf` → **HealthFacility**
- **RehabilitationStructure** → `subClassOf` → **HealthFacility**

Semantic Links (Object Properties)

These relationships connect the core entities to the common and contextual entities, establishing the domain context.

- **HealthFacility** → `locatedIn` → **Municipality**: Links a facility to its geographical territory

Purpose @ Territory KG (contd.)



Purpose @ Territory KG (contd.)

3.2 Information Gathering

In this section, an overview of the primary input data sources available for the project is provided. This includes a list of resources that span across languages, schemas, and data values.

3.2.1 Knowledge Layer

This section outlines the semantic resources identified and utilized to structure the Knowledge Graph. For this project, Schema.org was selected as the exclusive reference ontology. This decision was driven by the need for a lightweight, highly interoperable, and web-standard vocabulary that could sufficiently model both the healthcare and territorial aspects of the domain without the excessive complexity of clinical-specific standards like FHIR.

Schema.org Vocabulary provided the foundational class hierarchy and property definitions for the majority of the entities in the Knowledge Graph. Its broad coverage allowed for a consistent modeling approach across different domains:

- **Healthcare Facilities:** The $schema:MedicalOrganization$ class served as the semantic backbone for the core $HealthFacility$ entity. Specific facility types were mapped directly to their Schema.org equivalents, such as $schema:Hospital$ for hospitals and $schema:Pharmacy$ for dispensaries.
- **Territory and Location:** The $schema:AdministrativeArea$ class was used to model municipalities, providing a standard way to represent administrative boundaries. Furthermore, the $schema:Place$ and $schema:GeoCoordinates$ (using latitude and longitude properties) were utilized to model the physical location of every facility, ensuring compatibility with standard geospatial querying tools.
- **Organization and Attributes:** Common attributes found in the source datasets, such as addresses, tax IDs, and contact information, were mapped to standard Schema.org properties like $schema:address$, $schema:vatID$, and $schema:telephone$.

Purpose @ Territory KG (contd.)

3.2.2 Data Layer

Hospital Resource The Hospital Resource is a data value dataset that provides data on hospital facilities, specifically focusing on those within the Trentino region. This resource includes detailed administrative and location metrics for each facility, offering insights into the type of hospital management and contact information. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_OSP (int64):** The unique administrative code for the hospital.
- **OSPEDALE (string):** The official name of the hospital facility.
- **INDIRIZZO (string):** The street address of the facility.
- **COMUNE (string):** The municipality where the hospital is located.

- **TIPO_OSP (string):** Description of the hospital type (e.g., "OSPEDALE A GESTIONE DIRETTA").
- **TELEFONO (int64):** Contact phone number.
- **LATITUDINE_P / LONGITUDINE_P (float):** The geographic coordinates of the facility.

Pharmacy Resource The Pharmacy Resource is a data value dataset that provides information on pharmacies, specifically focusing on establishments within the Trentino region. This resource identifies pharmacies by their administrative codes and provides their precise location and operational start dates. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_FARMACIA (int64):** The unique administrative code for the pharmacy.
- **FARMACIA (string):** The name of the pharmacy.
- **TIPOLOGIA (string):** The classification of the pharmacy (e.g., "ORDINARIA").
- **DATA_INIZIO (string):** The date when the pharmacy commenced operations.
- **INDIRIZZO (string):** The street address.
- **LATITUDINE_P / LONGITUDINE_P (float):** The geographic coordinates.

Para-pharmacy Resource The Para-pharmacy Resource is a data value dataset that provides in-

Purpose @ Territory KG (contd.)

3.3 Report

3.3.1 Personas and Scenarios

We maintained the same core scenarios and personas throughout the process, refining them to better match the specific data attributes found (e.g., focusing on specific services like "Psychiatric Assistance" because that data was available).

3.3.2 Competency Questions (CQs)

Initially, our project used a set of preliminary scratch competency questions that were based on an imagined or ideal concept of the data we might have, rather than being drawn from actual information.

Once the final datasets were selected and confirmed for the project, we conducted a crucial review to ensure these questions matched the data's real content and limitations.

Adjustment: Questions were either reworded or entirely removed to be directly relevant and answerable using the chosen data.

1.
   - **Old:** "Which municipalities in the Val di Fassa are more than a 30-minute drive from the nearest Type 1 Emergency Room?"
   - **New (Dr. Elena Rossi 3.1):** "Which municipalities do not have any health structure offering Psychiatric Assistance?"
   - **Change:** The specific focus on "Val di Fassa" and "30-minute drive" was removed because the datasets did not support real-time travel time calculations. The question was broadened to a province-wide gap analysis based on available service attributes (ASSISTENZA).
2.
   - **Old:** "What specialized services (e.g., pediatrics, cardiology) are available at the primary hospital serving the remote communities of Val di Fassa?"
   - **New (Giulia Verdi 2.3):** "What specialized services are available at the health structures in a specific municipality?"

Purpose @ Territory KG (contd.)

3.
   - **Old:** "Which local clinics (Poliambulatori) within 10km of Riva del Garda are currently accepting new primary care patients?"
   - **New (Marco Bianchi 1.3):** "List all pharmacies within a 5km radius of a specific health facility."
   - **Change:** The original question was dropped because the data sources provided no information on "accepting new patients" (real-time status). It was replaced with a feasible proximity query for pharmacies, which is supported by the coordinate data.
4.
   - **Old:** "What is the closest Trauma Center to the road intersection at Passo Pordoi, and what is the calculated ETA via helicopter transport?"
   - **New (Marco Bianchi 1.1):** "What is the nearest hospital to a current GPS coordinates?"
   - **Change:** Calculation of "ETA via helicopter" was impossible with the static facility data. The question was simplified to a direct geodesic distance calculation ("nearest hospital") using the LATITUDINE/LONGITUDINE fields.
5.
   - **Old:** "Which nearby hospitals are confirmed to have a functional medical evacuation helicopter pad and an available surgical team right now?"
   - **New (Luca Verdi 4.3):** "How many health structures are managed directly by the health service in a specific municipality?"
   - **Change:** Real-time resource availability ("available surgical team right now") is not in the Open Data. The question was replaced with a structural/administrative query relevant to the Planner persona (Luca Verdi).
6.
   - **Old:** "Which Dialysis Centers within 50km of San Martino di Castrozza offer short-term appointment slots for visiting patients?"
   - **New (Luca Verdi 4.1):** "What is the total number of beds for elderly care in a municipality?"
   - **Change:** "Appointment slots" implies a booking system, which doesn't exist in the data. The question was shifted to querying capacity (NUM_POSTI), which is a hard metric provided in the Residential Care dataset.

This focused process yielded an improved and refined set of competency questions that are now more exact, useful, and clearly supported by the information in our final data sources.

iTelos - Purpose Definition

Fausto Giunchiglia, Mayukh Bagchi

Lecture 15 @ KG 2026